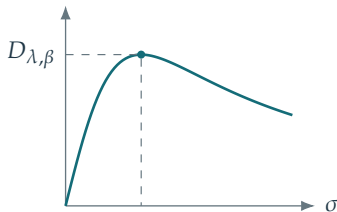


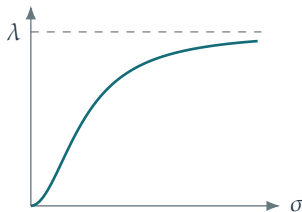
$$b_{\lambda,\beta}(\sigma) = \frac{\lambda\sigma^\beta}{\lambda + \sigma^2}, \quad D_{\lambda,\beta} = \sup_{\sigma \geq 0} b_{\lambda,\beta}(\sigma)$$

$$0 < \beta < 2$$



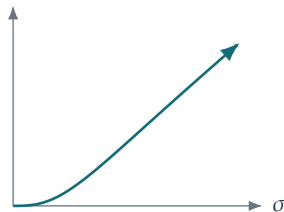
$$\sigma_* = \sqrt{\frac{\beta\lambda}{2-\beta}}$$

$$\beta = 2$$



$$D_{\lambda,2} = \lambda$$

$$\beta > 2$$



$$D_{\lambda,\beta} = +\infty$$

Here $\lambda > 0$ and the supremum ranges over all $\sigma \geq 0$.
On a fixed bounded operator, only $0 \leq \sigma \leq \|\Phi\|$ is relevant.